

© Zahra Shakeri, Dalla Lana School of Public Health, University of Toronto



REINFORCEMENT LEARNING

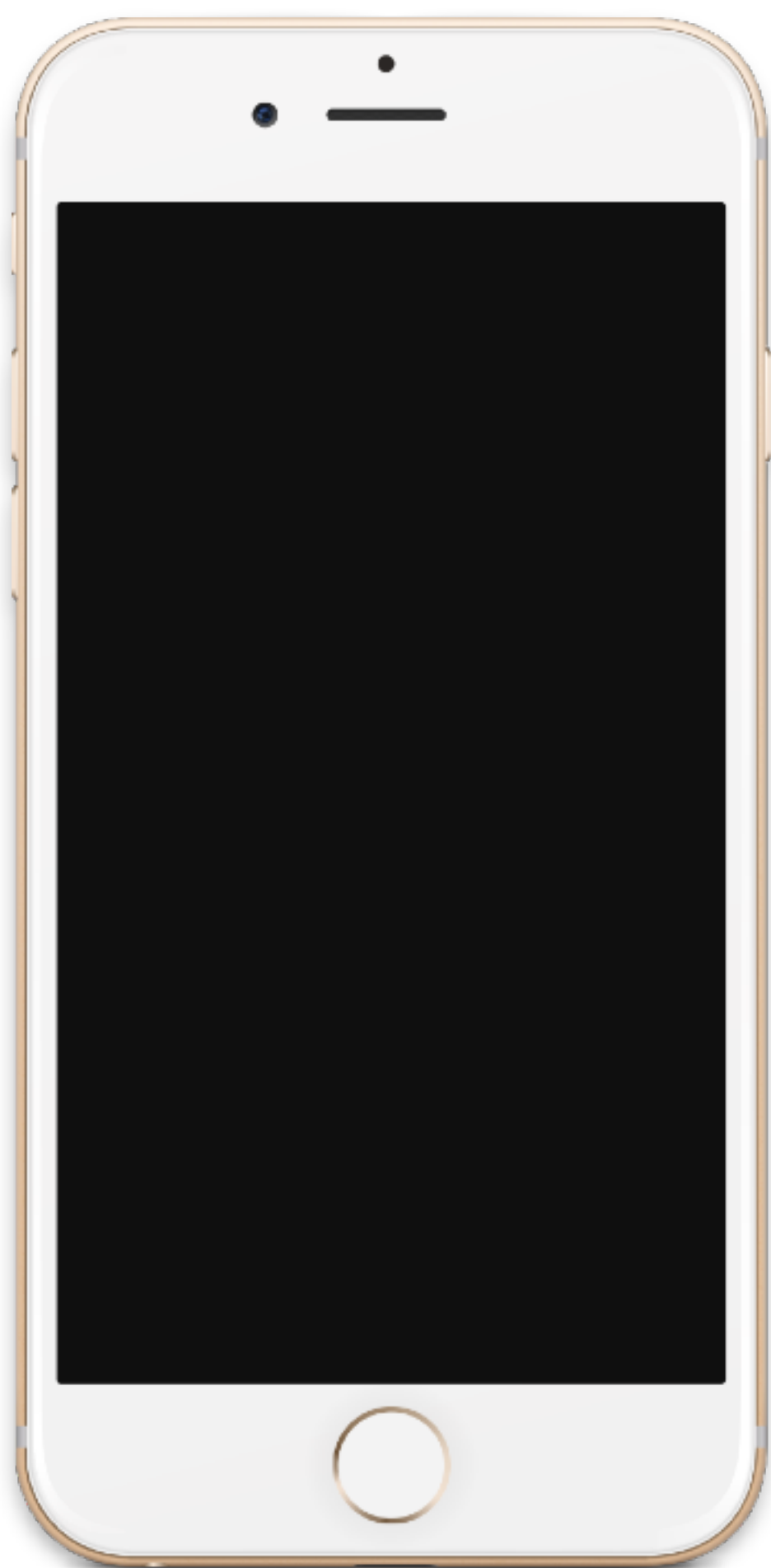
Teaching towards better decisions!

Reinforcement learning involves several key components: an agent, an environment, a set of states $\mathcal{S}$, a set of actions $\mathcal{A}$, a reward function $R(s, a)$, and a policy $\pi(a | s)$, which is a probability distribution over actions given states.

- ▶ Agent: The learner and decision maker.
- ▶ Environment: The external system the agent interacts with.
- ▶ State s : A situation the agent perceives.
- ▶ Action a : A move the agent makes.
- ▶ Reward R : A real number indicating the immediate benefit of performing action a in state s .
- ▶ Policy π : A mapping from states to actions, representing the agent's strategy.

The agent interacts with the environment in a series of discrete time steps. At each time step t , the agent receives a representation of the environment's state s_t , selects an action a_t , and sends this action to the environment. The environment responds by presenting a new state s_{t+1} to the agent and returning a reward r_t .

The agent's objective is to maximize the expected cumulative reward over time!









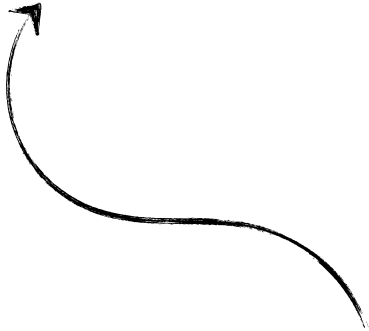




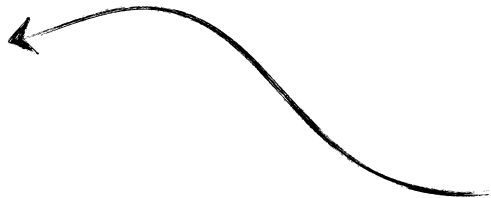


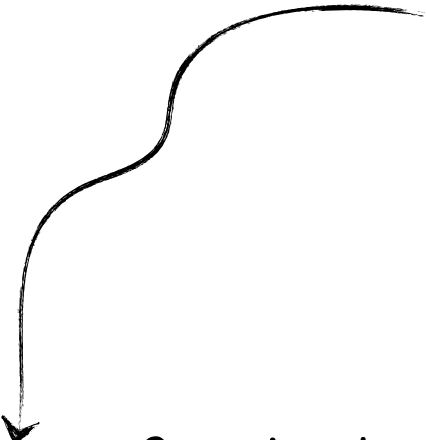


State: Current time and water intake (e.g. 9:30 am, 500ML).



Action: Sending or not
sending a reminder.





Rewards: +1 for drinking after
reminder, -1 for not, 0 for no
reminder.

Policy: Remind every
2 hours if < 250 ml
water in 2 hours.





4

9

REINFORCEMENT LEARNING

Teaching through rewards for better decisions!



Reinforcement learning involves several key components: an agent, an environment, a set of states S , a set of actions A , a reward function $R(s, a)$, and a policy $\pi(a | s)$, which is a probability distribution over actions given states.

- ▶ Agent: The learner and decision maker.
- ▶ Environment: The external system the agent interacts with.
- ▶ State s : A situation the agent perceives.
- ▶ Action a : A move the agent makes.
- ▶ Reward R : A real number indicating the immediate benefit of performing action a in state s .
- ▶ Policy π : A mapping from states to actions, representing the agent's strategy.

The agent interacts with the environment in a series of discrete time steps. At each time step t , the agent receives a representation of the environment's state s_t , selects an action a_t , and sends this action to the environment. The environment responds by presenting a new state s_{t+1} to the agent and returning a reward r_t .

The agent's objective is to maximize the expected cumulative reward over time!

State: Current time and water intake (e.g. 9:30 am, 500ML).

Policy: Remind every 2 hours if < 250 ml water in 2 hours.

Action: Sending or not sending a reminder.

Rewards: +1 for drinking after reminder, -1 for not, 0 for no reminder.



REINFORCEMENT LEARNING

Teaching through rewards for better decisions!



- ▶ In **supervised learning**, the data collection process involves collecting pairs of inputs and corresponding ground truth outputs. These pairs are used to train the model so that it can learn the mapping from inputs to outputs.
- ▶ In **unsupervised learning**, the data collection process involves collecting only the input data, as there are no corresponding ground truth outputs.
- ▶ In **reinforcement learning**, the data collection process involves the agent interacting with the environment and collecting states, actions, and rewards. This data is then used to train the model to learn a policy that maximizes the expected cumulative reward.